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Frequency and Type of Hedging Devices Used in the Research Articles of Humanities, Basic Sciences and Agriculture

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Abstract

The present study compared and contrasted the frequency of use of hedging devices used in the research articles of three fields of Humanities, Basic Sciences and Agriculture and three sub-fields within each field (i.e. Law, Persian literature and TEFL within Humanities; Biotechnology, Soil Sciences and Horticulture within Agriculture; and Biology, Applied Chemistry and Geology within Basic Sciences). In so doing, 180 research articles, 20 from each sub-field, were selected and analyzed based on Salager-Meyer's (1994) taxonomy of hedging. The results of the chi-square analyses revealed, regarding the frequency of use and type of hedging devices, there were significant differences among: a) the three fields of Humanities, Basic Sciences and Agriculture; b) the sub-fields of Humanities, Basic Sciences and Agriculture.

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1. Introduction

Writing is considered as the most crucial and permanent way of communication, which has several forms, the most important of which is academic writing. Academic writing involves interrelationships between the author and their audiences based on certain conventions, which take part in an academic discourse. These conventions ensure

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the discourse community that the readers and colleagues in the field accept the product.
A vital issue in academic writing, especially in writing research articles, which is regarded as a difficult issue for non-native writers, is the use of so called hedging devices. Hedging was first used by Lakoff (1973) as "words whose job is to make things more or less fuzzy" (p. 195) and has been applied to words such as 'might', 'perhaps', 'maybe', 'I think', etc which are used to show uncertainty. Hedging is an important aspect of linguistic features of academic genres, and research articles have been regarded as the main corpus for studying hedging devices. The main use of hedging is to talk about the findings of the study in a cautious manner while allowing the readers to have different interpretations. Exactness is the inevitable core of scientific writing. Many believe that a scientific writer is supposed to transfer the information in a direct and explicit way. However, there are many reasons for a scientific writer not to do so, for example, he may prefer not to be exact about a scientific claim because of its consequences or not being completely sure. Brown and Levinson (1988), believe that being exact in scientific claims is very threatening. Even if the writer is totally sure about their claims, they may prefer not to utter it in an exact and precise way, so that the writer can leave some room for error.

1.1. Hedging

Hedging is a linguistic device, which has multiple objectives; its adoption will help the authors and researchers to express their claims with due care and preciseness. Like all the other scientific notions, there are different definitions for the term "hedging". That is, different authors have developed different definitions, taxonomies and functions for hedging.

The pioneer of hedging is perhaps George Lakoff , who termed the concept of cautious language as "hedging" and defined it as "words or phrases whose job is to make things more or less fuzzy" (1972, p. 471). Hedging, in fact, gives the authors the opportunity to be cautious about expressing claims, which are not proven.

Salager-Meyer (1994) believes there are two purposes for using hedging devices: 1. Making issues fuzzy. Salager-Meyer holds the view that if the author presents the material explicitly, the chance of criticism will be given to other authors. 2. Using-hedging devices, the authors can enhance the precisions of their claims. Salager-Meyer (1994, p.151) maintains, "Hedging may present … the strongest claim a careful researcher can make".

The purpose of this study was to compare and contrast the hedging devices used in nine sub-fields within three different fields. The sub-fields selected were Law, Persian Literature and TEFL within Humanities; Biotechnology, Soil Sciences and Horticulture within Agriculture; Biology, Applied Chemistry and Geology within Basic Sciences. The hedging devices used in the Discussion section of the articles published in these fields and sub-fields published between 2000 and 2012 were analyzed, compared and contrasted.

1.2. Research Questions

The following research questions were formulated for the current study:

1. Is there any significant difference in the frequency of use of hedges in the Discussion section of articles of the three fields of Humanities, Basic Sciences and Agriculture?
2. Is there any significant difference in the frequency of use of hedges in the Discussion section of articles of the three selected sub-fields of Persian Literature, Law, and TEFL within Humanities?
3. Is there any significant difference in the frequency of use of hedges in the Discussion section of articles of the three selected sub-fields of Applied Chemistry, Biology and Geology within Basic Sciences?
4. Is there any significant difference in the frequency of use of hedges in the Discussion section of articles of the three selected sub-fields of Biotechnology, Soil Sciences and Horticulture within Agriculture?

2. Methodology

2.1. Materials

The corpus of the study included 180 English articles (published between 2000 and 2012) which were chosen

based on random selection. Nine different sub-fields from three different fields were included in the study. The sub-fields were Law, Persian Literature and TEFL within Humanities; Biotechnology, Soil Sciences and Horticulture within Agriculture; and Biology, Applied Chemistry and Geology within Basic Sciences. 60 articles were chosen for each main field (20 for each sub-field).

2.2. Procedure

In order to identify the percentage and frequency of use of hedging devices in each field and sub-field, the Discussion sections of all the articles were read meticulously word by word by the researchers and the hedging devices used in each article were counted. The hedging devices were identified based on Salager-Meyer's (1994) taxonomy which divides hedges into five parts of *shields* (e.g. can, may, etc), *approximators* (e.g. about, occasionally, etc), *authors' personal doubt and direct involvement* (e.g. I believe), *emotionally-charged intensifiers* (e.g. extremely, unexpectedly, etc) and finally *compound hedges* (e.g. would seem somewhat).

2.3. Data Analysis

In order to analyze the data, Chi-square analyses were run. The alpha value was set at .05. The frequency of hedging devices used was calculated. Inter-rater reliability coefficient for extracting and counting the frequency of hedging was calculated to be .89.

3. Results

The frequency and type of hedging devices of all the sub-fields were identified which are shown in Table 1.

Table 1. Descriptive statistics for hedging devices in different sub-fields

	Persian Literature	**Law**	**TEFL**	**Applied Chemistry**	**Biology**	**Geology**	**Bio-technology**	**Soil Sciences**	**Horticulture**	**Tl**
Shield	N=115 P=30.3%	N=111 P=27.1 %	N=147 P=36 %	N=143 P=42%	N=100 P=49%	N=129 P=51%	N=54 P=37%	N=142 P=43%	N=120 P=48%	10 61
Approxr.	N=101 P=26.6%	N=70 P=17%	N=98 P=24 %	N=105 P=31%	N=67 P=33%	N=62 P=24.2 %	N=38 P=26%	N=89 P=27%	N=82 P=33%	71 2
Authors Doubt	N=28 P=7.3%	N=58 P=14.2 %	N=29 P=7%	N=21 P=6.3%	N=7 P=3.2%	N=9 P=3.4%	N=14 P=9.3%	N=27 P=8.2%	N=19 P=7.6%	21 2
Emotion. Charged Intensifs	N=92 P=24.1%	N=86 P=21%	N=94 P=23 %	N=24 P=7%	N=23 P=11.4 %	N=43 P=16.7 %	N=25 P=17%	N=39 P=12%	N=17 P=7%	44 3
Compound Hedges	N=44 P=11.7%	N=85 P=20.7 %	N=40 P=10 %	N=47 P=13.7%	N=7 P=3.6%	N=12 P=4.7%	N=16 P=10.7%	N=32 P=9.8%	N=11 P=4.4%	29 4
Total	380	410	408	340	204	255	147	329	249	

The first research question

To investigate the first question of the present study, a chi-square analysis was run whose results show there is a

statistically significant difference between the three fields of Humanities, Basic Sciences and Agriculture regarding the frequency of hedges used.

Table2. Chi-square analysis for overall difference among the three fields regarding their use of hedging devices

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	1.035E2[a]	4	.000
Likelihood Ratio	106.703	4	.000
N of Valid Cases	1997		

The second research question

The second research question investigated the existence of any difference among the three sub-fields of Humanities (i.e. Persian Literature, Law, TEFL). Chi-square analyses were run whose results revealed that there existed a statistically significant difference among the sub-fields of Humanities regarding the frequency of hedges used.

Table3. Chi-square analysis for overall difference among the sub- fields of Humanities regarding their use of hedging devices.

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	48.778[a]	8	.000
Likelihood Ratio	47.781	8	.000
N of Valid Cases	1197		

The third research question

Another chi-square was run to investigate the differences among the sub-fields of Basic Sciences (i.e. Applied Chemistry, Biology, and Geology) whose results revealed there was a statistically significant difference among the sub-fields of the Basic Sciences regarding the frequency of use of hedging devices.

Table 4. Chi-square analysis for overall difference among the sub-fields of Basic Sciences regarding their use of hedging devices

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	43.851[a]	8	.000
Likelihood Ratio	43.999	8	.000
N of Valid Cases	799		

The fourth research question

Another chi-square analysis was run to compare the three sub-fields of Agriculture.

The results revealed the three sub-fields of Biotechnology, Soil Sciences and Horticulture were also significantly different with regard to the frequency of hedging device used.

Table5. Chi-square analysis for overall difference among the sub-fields of Agriculture regarding their use of hedging devices

	Value	df	Asymp. Sig. (2-sided)
Pearson Chi-Square	20.921[a]	8	.007
Likelihood Ratio	21.639	8	.006
N of Valid Cases	725		

4. Discussion

The analysis of the Discussion sections of the articles of the three fields of Humanities, Basic Sciences and Agriculture revealed there were statistically significant differences among them regarding their frequency of use of hedging devices.

The results of analyzing the Discussion section of the articles of the three fields indicated Humanities article writers were more tentative in putting forward their claims and in rejecting or confirming the ideas of the others; that is to say, they used more hedges than the writers of the other two fields; the total frequency of all types of hedging devices which were used in Humanities was 1198.

On the other hand, the results of the study indicated that the overall frequency of the hedging devices used in Basic Sciences and Agriculture fields were 799 and 725 respectively which is relatively lower than that used in the field of Humanities.

The Basic Sciences and Agriculture article writers seemed not to hedge their statements and were relatively more assertive and certain in their claims. In our view, one possible reason for the emergence of this result could be

attributed to the fact that the Humanities fields are not exact in nature, but instead deal with implicit and untouchable matters which enforce the authors to be more cautious and uncertain in and about their claims (i.e. to use more hedging devices) whereas Basic Sciences and Agriculture deal with rather exact matters, possibly numbers, and experiments which leave out no room for the use of too much hedging and tentativeness.

The results of the analysis of the three fields revealed that hedges of the types *shields, approximators and emotionally charged intensifiers* were the most frequently used hedges employed in the Discussion section of research articles.

This finding is, to some degree, in line with those of Salager-Meyer (1994) who found that the three most frequently used hedging devices in research articles were *shields, approximators and compound hedges* which accounted for over 90% of the total number of hedges used in research articles.

However, in the present study, the third most frequently used hedging device was found to be "*emotionally charged intensifiers"* which is in contrast to that of Salager-Meyer (1994) who found *compound hedges* to be the third most frequently used type. Salager-Meyer (1994) claims that the three most frequently used hedges in the study accounted for over 90% of the hedges used, while in our study, the three most frequently used hedges accounted for 81.41% of the total hedges used.

The most frequently used type of hedging devices in the present study was found to be shields (38.97% of the total hedges used). This finding is in parallel with the results of the studies conduct by Kubui (1988, as cited in Salager-Meyer, 1994), Bonyadi, Gholami and Nasiri (2012), who found that the most frequently used hedging devices in academic writings were *shields.*

The findings of the study might be of assistance to ESP/EAP curriculum developers, syllabus designers, and research writers.

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